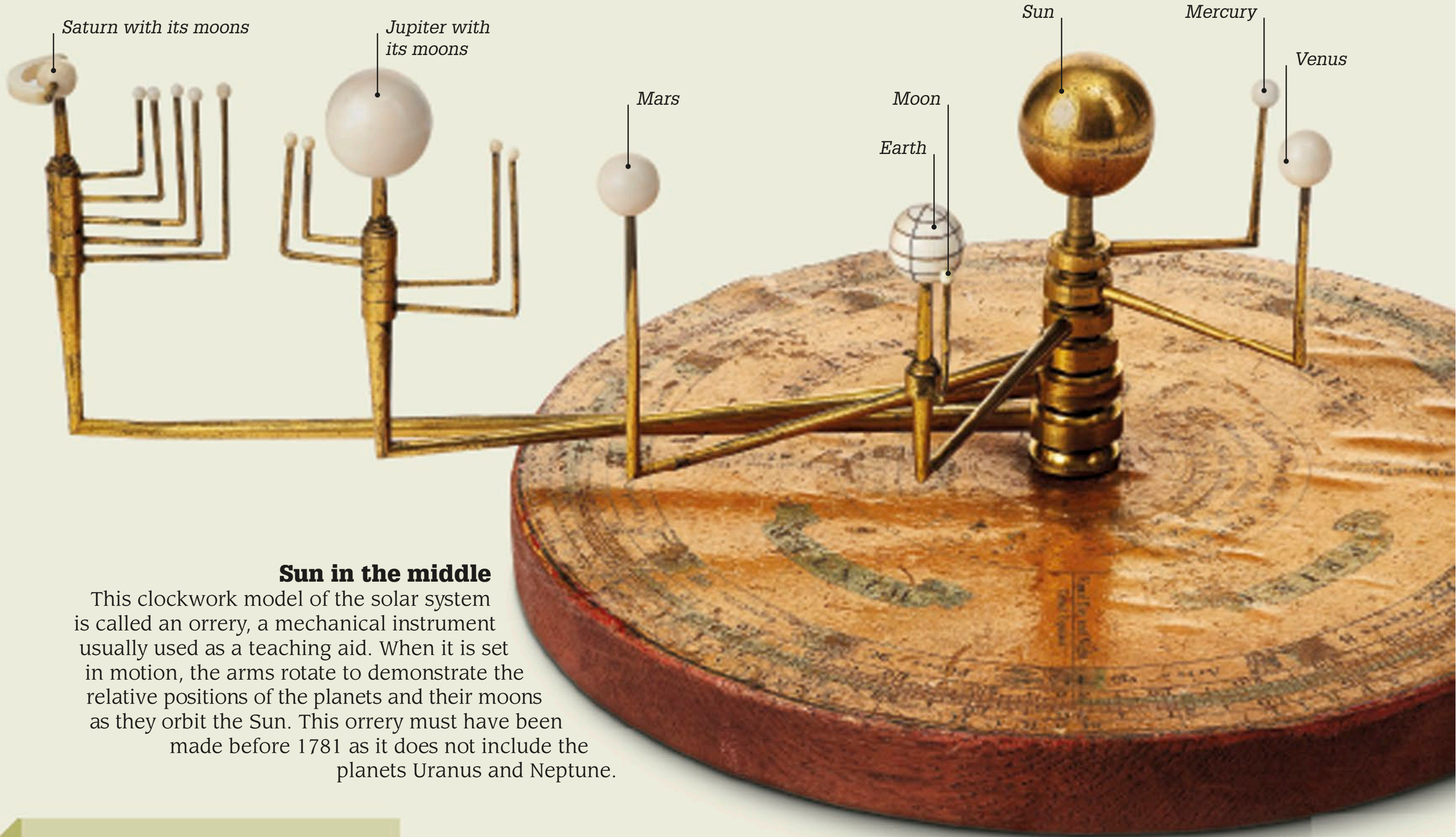
Circular orbits
Nicolaus Copernicus was the first European astronomer to argue that the Sun was at the center of the Universe. He believed that Earth and the other planets traveled around it in circular orbits. He published his revolutionary ideas just before his death in 1543.

Copernicus's drawing of the Sun (at the center) and planets

Paths in the sky

Astronomers had no idea what keeps planets in their orbits, until English scientist Isaac Newton realized that it is a force called gravity. This force, which makes objects fall down on Earth, is the same force that keeps the planets from flying off in straight lines. The planets are actually falling toward the Sun—but they are also moving sideways in their orbits. If they stopped moving, they would crash into the Sun.



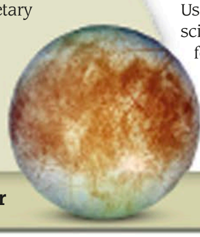
Sun in the middle
This clockwork model of the solar system is called an orrery, a mechanical instrument usually used as a teaching aid. When it is set in motion, the arms rotate to demonstrate the relative positions of the planets and their moons as they orbit the Sun. This orrery must have been made before 1781 as it does not include the planets Uranus and Neptune.

Key events

140 CE
Greek mathematician Ptolemy stated that Earth is the fixed center of the Universe. His views were not challenged for the next 1,500 years.

1543
Nicolaus Copernicus published a book in which he proposed that Earth and the other planets travel around the Sun in circular orbits.

1609
In his three laws of planetary motion, Johannes Kepler proved mathematically that the planets travel in elliptical paths.



Europa, a moon of Jupiter

1610
Using a telescope, Italian scientist Galileo observed four moons in orbit around Jupiter, proving that not everything in space orbits Earth.